



Manufactured by Thawzall,

LLC 104 Donovan Drive

P O Box 158

Alexandria, MN 56308,
USA

T 320.759.1588 24/7 Tech Support 888.757.3545

F 320.759.1583

www.thawzall.com E-mail: thawzall@thawzall.com

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For model:

H750

Please record the following information from your new Thawzall for future reference. This information is required for all warranty claims.

Purchase date: _____

Machine model: _____

Machine serial number: _____

Serial number located on trailer pull tube.

Thawzall, LLC

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Donovan Drive

Alexandria, MN 56308

Phone: 320.759.1588

Fax: 320.759.1583

Tech Support: 888.757.3545

Website: www.Thawzall.com

E-Mail: Thawzall@thawzall.com

MISSION

COMPANY GENERAL INFORMATION

To provide fuel efficient, rental-store friendly portable heaters and provide 24/7 technical support via our toll free number: (888) 757-3545

VISION

To become a world class manufacturing company where our employees feel proud of their achievements

VALUES

Customers (internal and external) feel honored, management is humbly grateful and Thawzall has made a positive difference

We take great pride with every machine we build!

Thank you for investing in a Thawzall portable heater!

IMPORTANT SAFETY RECOMMENDATIONS AND WARNINGS

OUTSIDE SAFETY FEATURES

Your Thawzall is equipped with DOT Certified outside LED lights, reflectors, safety-chains, and electronic break away brakes

Make use of these features:

Before towing the trailer, be sure that all the lighting is connected and working. Connect the safety chains to the towing vehicle.

Connect the small cable from the break-away switch to the towing vehicle.

PARKING YOUR THAWZALL

Avoid unexpected movement of your Thawzall: Avoid parking on hills.

Use blocks or wheel chocks to prevent movement. Always use the jack to support the hitch.

TOWING YOUR THAWZALL

Tow safely:

A $\frac{3}{4}$ ton truck with a brake controller is recommended for towing the H750.

Thawzall axles, wheels and tires are rated to travel at legal posted speeds on the highway. Connect safety-chains, lights and brake cable to towing vehicle.

Thawzall trailers are equipped with electronic break-away brakes in case the trailer should become disconnected from the towing vehicle. Trailer brakes will apply automatically.

Failure to connect trailer in a safe manner could result in a serious accident or death.

Always verify that the hitch ball size on the towing vehicle matches the size of the coupler on the Thawzall trailer.

SUPPORTING DOCUMENTS

Operation and Maintenance Manual Cummins QSB5 Engine, DSFAE Generator

Owner Manual, Installation/Operation Controller 1302, Cummins

PRE OPERATING INSTRUCTIONS

Do not operate your Thawzall without instruction and understanding of the startup and shut down procedures thoroughly—If you do not understand these instructions, call Thawzall Technical Support—888.757.3545

Keep your machine in proper working condition. Unauthorized modifications to the machine may impair the function and/or the safety and affect the life of the machine.

GFI Outlet. Always plug the electrical cord for the space heater from your Thawzall into a GFI protected outlet.

Failure to do so could cause shock or electrocution.

Safety Glasses. Always wear safety glasses when operating your Thawzall, especially when fueling the diesel engine or maintaining the battery.

Caution and warning decals. Carefully observe and maintain all of the caution and warning decals placed on your Thawzall machine. They are there to ensure proper operation of the Thawzall and your safety! Never “Hotwire” any of the electrical wiring on your Thawzall machine. Wiring circuits are carefully designed to provide for safe Startup and operation of the furnace, pumps and hose reel. Any alteration may cause an unsafe condition and could cause any or all components to malfunction or operate out of sequence.

Operate this machine only OUTDOORS! Never operate engine inside a building without proper venting of the exhaust to the outside. Carbon monoxide poisoning, asphyxiation and/or death may occur if exhaust is inhaled.

HANDLE DIESEL FUEL FOR ENGINE SAFELY

Do not refuel the machine while smoking or while an open flame or sparks are near the fueling point! Do not refuel while the engine and generator are running.

A clean machine is a safe machine. Prevent fires by keeping your Thawzall clean of accumulated trash, grease and debris. Clean up spilled fuel for the diesel engine properly.

CONTROLS AND INDICATORS

Provided in this section is the location and function of all the controls and indicators.

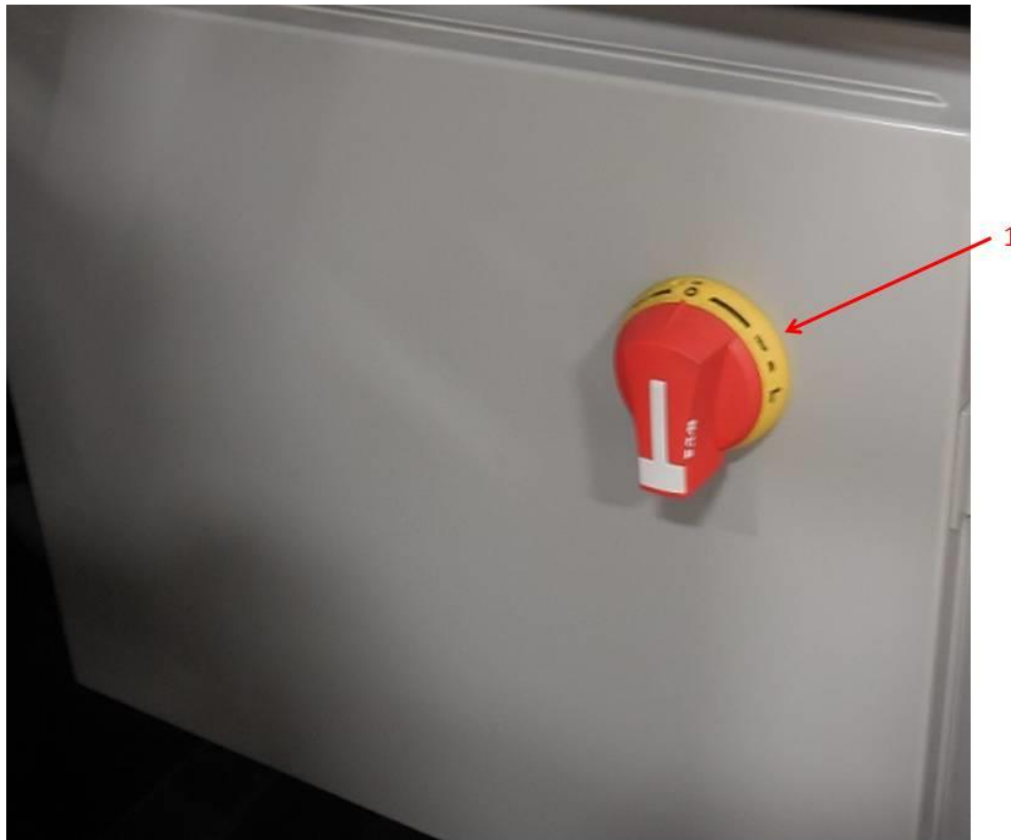


Figure: Controls On Outside 480 VAC Power Distribution Panel

Table: Controls On Outside Of 480 VAC Power Distribution Panel		
Figure Index No.	Control or Indicator	Function
1	Main Disconnect circuit breaker MD0010	Main 480 VAC 3 phase power disconnect from system

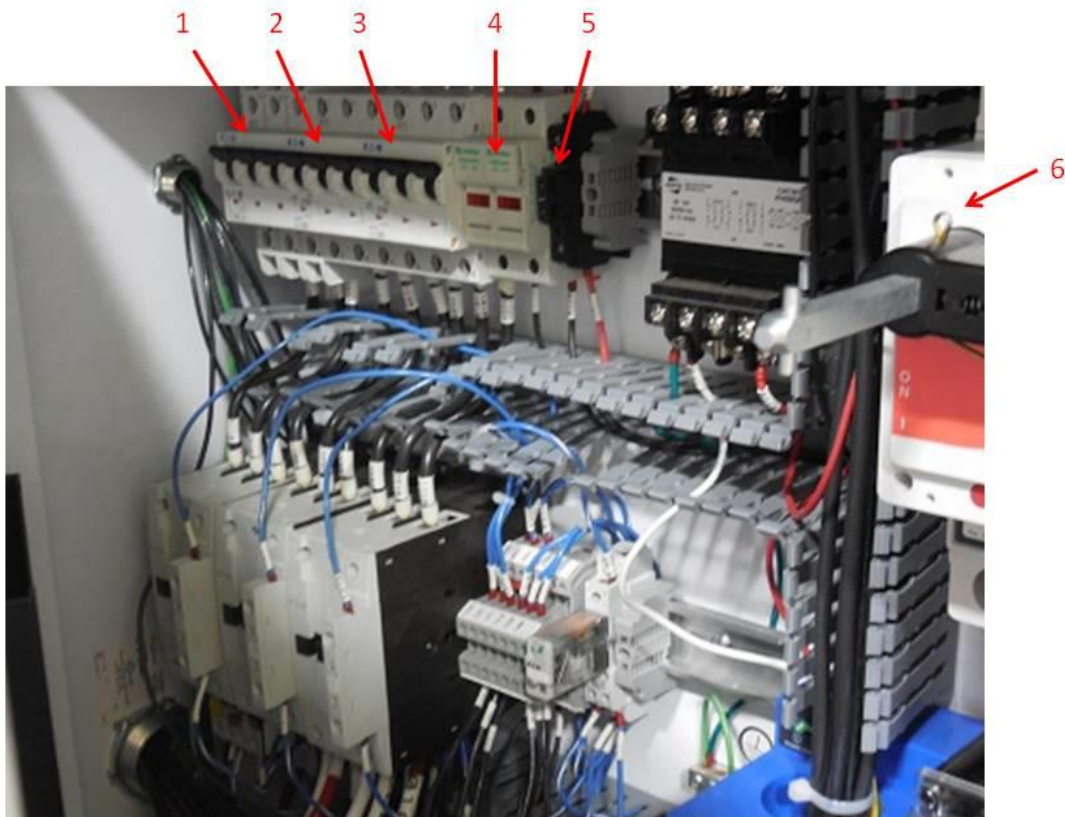


Figure: Controls Inside 480 VAC Power Distribution Panel

Table: Controls Inside 480 VAC Power Distribution Panel		
Figure Index No.	Control or Indicator	Function
1	Circuit breaker CB-0010	Power to fan
2	Circuit breaker CB-0090	Power to heater banks 1 and 2
3	Circuit breaker CB-0170	Power to heater banks 3 and 4
4	Fuses F0110 and F0130	Fuses for GFI module transformer on primary side
5	Fuse F0120	Fuse for GFI module transformer on secondary side
6	Main Disconnect circuit breaker MD0010	Main 480 VAC 3phase power disconnect from system.



Figure: Controls And Indicators Outside The Operator's Control Panel

Table: Controls And Indicators Outside The Operator's Control Panel		
Figure Index No.	Control or Indicator	Function
1	Lockout key	Used to lockout and prevent unauthorized starting of the unit.
2	Cummins HMI display	Used to start engine, stop engine, and monitor status
3	Fuel gauge	Monitors amount of diesel fuel in tank for engine
4	EMERGENCY STOP switch	When pushed, shuts unit down FOR EMERGENCIES ONLY
5	HEATER POWER switch	LOW – Only heater banks 1 and 2 turned on. HIGH – All four heater banks turned on.
6	DOOR INTERLOCK	OFF – Door interlocks enabled. ON – Door interlocks disabled.

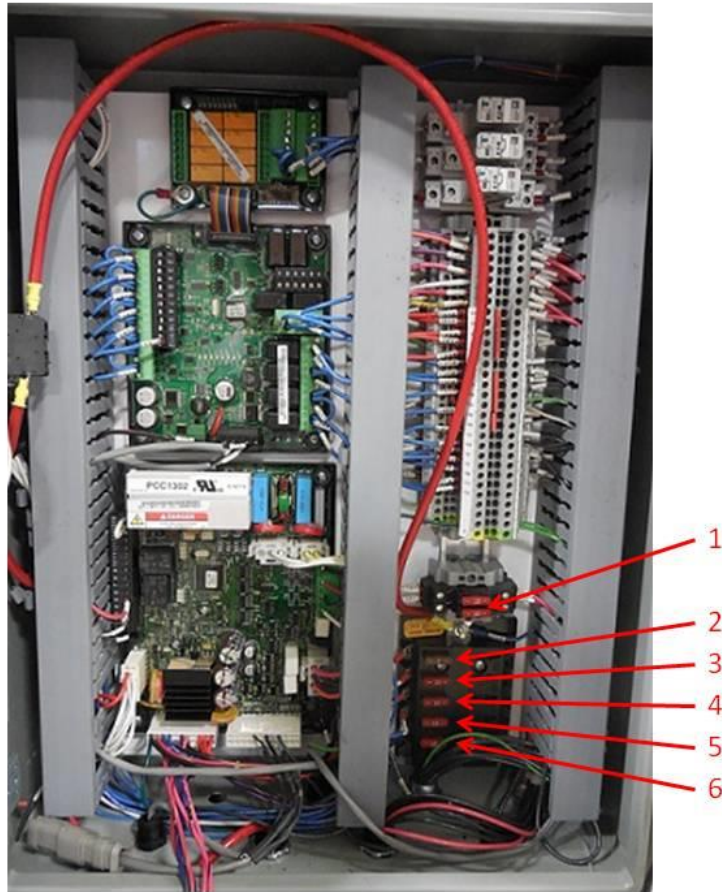


Figure: Controls and Indicators Inside Operator's Control Panel

Table: Controls and Indicators Inside Operator's Control Panel		
Figure Index No.	Control or Indicator	Function
1	Fuse 1180	Power to PMG
2	Fuse 630	Power to ECM
3	Fuse 1200	Power to Cummins Aux 102
4	Fuse 1120	Power to Cummins 211
5	Fuse 1040	Power to Cummins PCC 1302
6	Spare	Spare



NEVER SHUT THE ENGINE DOWN USING THE BATTERY DISCONNECT!!—This will destroy the program and the unit will not run

Figure: Battery Disconnect Control

Table: Battery Disconnect Control		
Figure Index No.	Control or Indicator	Function
1	Battery Disconnect switch	Applies battery power to the system. HMI display is on when power is applied.
CAUTION Battery isolation switch function is as follows: - Prevent battery discharge during vehicle shipping or storage - Protect service technicians from the danger caused by inadvertent engine crank. To offer good protection of service personnel. The BATTERY DISCONNECT switch is not suitable method for stopping the engine, as it is not guaranteed to stop the engine as the ECU may continue to operate with power generated by the alternator. It is also possible that opening the BATTERY DISCONNECT switch when engine is running could cause an alternator load dump and damage the electronic components.		

STARTING PROCEDURE

1. Check diesel fuel tank for fuel as shown in Figure: Start Procedure.
2. Make sure E-STOP button on Operator's Control Panel is pulled out and not activated.
3. Make sure DOOR INTERLOCK switch is set to OFF and access doors are closed.
4. Set HEATER POWER switch to LOW.>
5. Turn key switch on located on side of Operator's Control Panel to on.
6. Set BATTERY DISCONNECT switch to ON to apply electrical power to the system and the HMI display on the Operator's Control Panel shall display the First HMI Screen.
7. Select and press the button located below the Hand Icon (second from left) on the First HMI Screen on the Operator's Control Panel to access the second screen used to start the engine and generator.
8. Select and press the button located below the Hand Icon (second from right) on the Second HMI Screen. Engine will start and idle for a short period of time until warmed up before the generator shall automatically start. System is now operational.

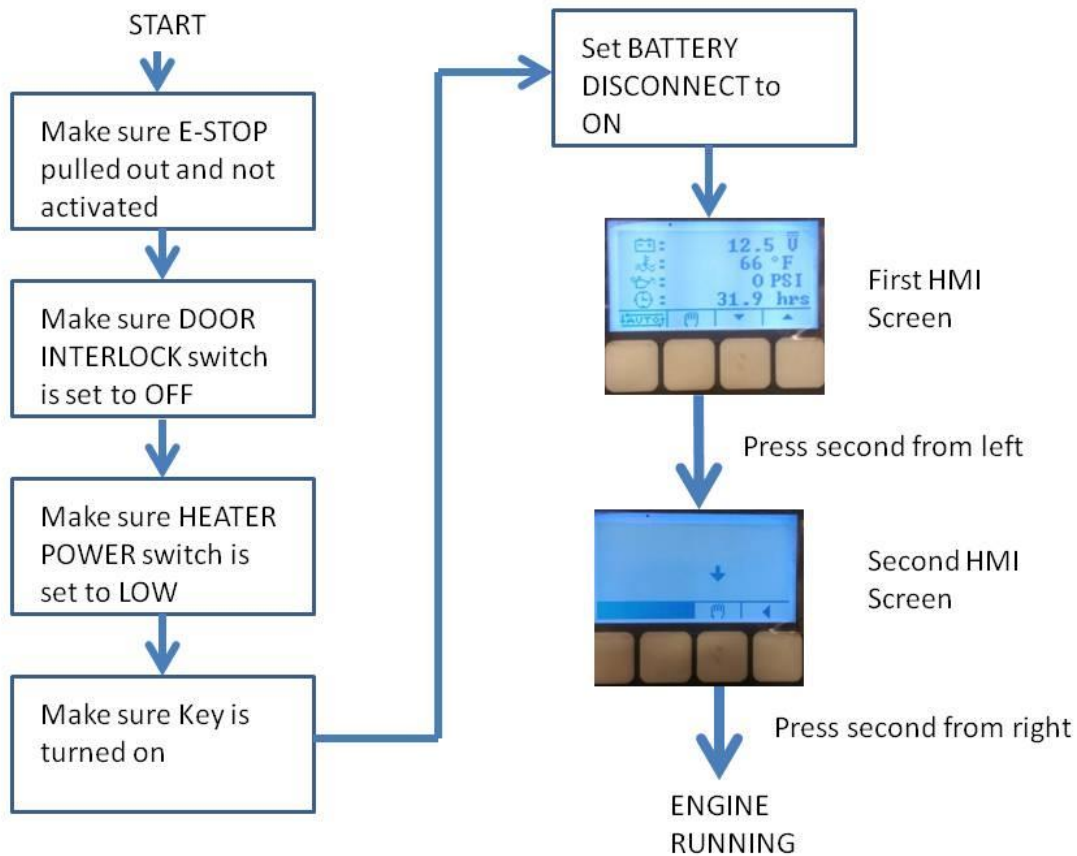


Figure: Startup Procedure

OPERATING PROCEDURES

1. Make sure power switch on 480 VAC Power Distribution Panel is turned on.
2. Select either LOW or HIGH on the HEATER POWER switch on the Operator's Control Panel to select the operating temperature of the output heated air.
3. Use the chart shown in Figure: Operator Screens to select which buttons to press to observe the different status screens for the system.
4. Check status as required.

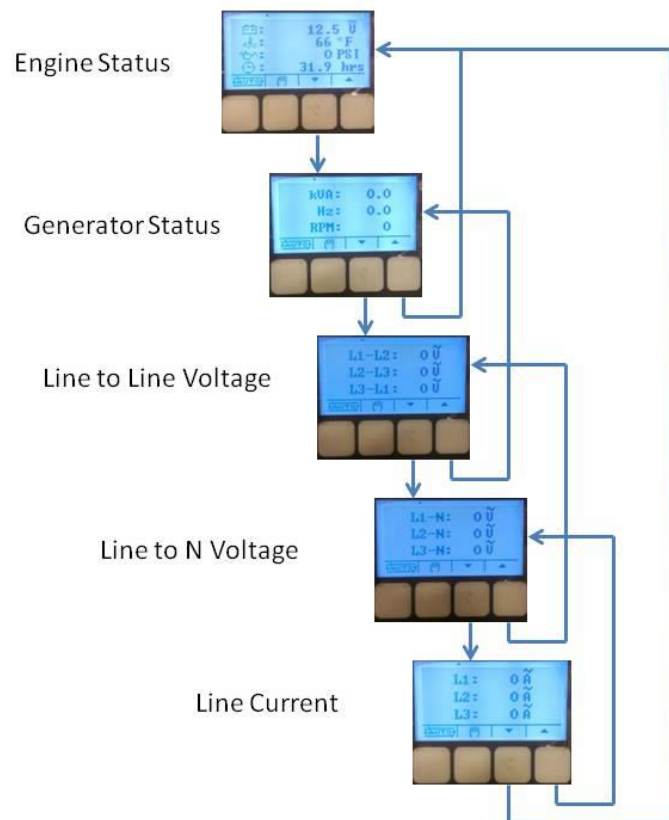
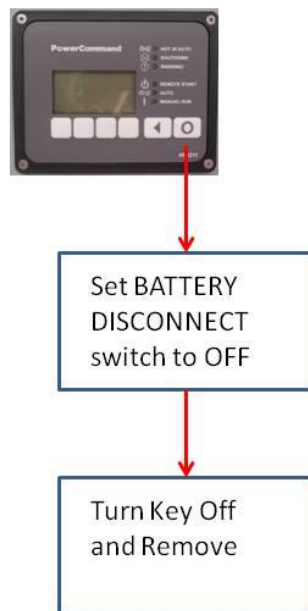


Figure: Operator Screens

SHUTDOWN PROCEDURE

1. Select and press the O button on the HMI display on the Operator's Control Panel regardless of which status screen is currently displayed to shut the engine down as shown in Figure: Shutdown Procedure. Engine shall turn off.
2. Set BATTERY DISCONNECT switch to OFF—**ONLY AFTER THE “O” button has been pressed and the engine is shut down and ECM save has finished**—to disconnect battery from system. The system is completely turned off.
3. Turn key to off and remove to prevent unauthorized use of unit.



EMERGENCY STOP PROCEDURE

1. In an emergency, press the E-STOP button on the Operator's Control Panel to turn off the system.
2. Investigate emergency and take appropriate action.

THEORY OF OPERATION

Referring to Schematic 82046 the H750 contains the engine, 480 VAC generator, engine coolant level sensor, engine control module (ECM), engine coolant level sensor, diesel fuel level sensor, generator control panel, heater duct over temperature sensor, 480 VAC power distribution panel, operator's control panel, and status lights.

THEORY OF OPERATION FOR 480 VAC POWER DISTRIBUTION PANEL

The 480 VAC power distribution panel controls the application of electrical power to the heater banks and air fan motor. It contains safety equipment (circuit breakers, ground fault interrupt (GFI), fuses, and system control equipment (contactors and starter) activated from the operator's control panel. These features are shown in the Figure: 480 VAC Power Distribution Panel Theory.

The diesel engine provides the original source of power for the system. The diesel engine drives a 480 VAC, 3 phase, wye connected generator. The middle of the wye is connected to the chassis. The three phases of the generator feeds the main disconnect circuit breaker MD0010 on the power distribution panel. The purpose of the main disconnect circuit breaker is to lock out all 480 VAC power without shutting down the diesel engine. It is not to be used for activation while energized.

When the main disconnect circuit breaker MD0010 is activated three phase power is routed through the GFI module to provide safety monitoring. The GFI module senses differences in line current between the three phases. A difference of more than 30 mA causes the GFI module to shut down the 480 VAC power. The GFI module is powered by a transformer T-0120.

Once the three phase power is routed through the GFI, it is distributed to three circuit breaker branches CB-0010, CB-0090, and CB-0170. The circuit breaker CB-0010 branch also contains contactor C-1250 connected in series to the output of the circuit breaker. The output of contactor C-1250 is connected to heater banks 1 and 2 the final stage of the circuit. Contactor C-1250 is controlled by a signal from the operator's control panel.

The circuit breaker CB-0090 branch also contains contactor C-1270 connected in series to the output of the circuit breaker. The output of contactor C-1270 is connected to heater banks 3 and 4 the final stage of the circuit. Contactor C-1270 is controlled by a signal from the operator's control panel.

The circuit breaker CB-0170 branch also contains contactor MS-1230 connected in series to the output of the circuit breaker. The output of contactor MS-1230 is connected to fan motor the final stage of the circuit. Contactor MS-1230 is controlled by a signal from the operator's control panel.

Only heater banks 1 and 2 are on when the HEATER POWER switch on the operator's control panel is set to LOW. All four heater banks are on when the HEATER POWER switch is set to HIGH. The fan motor is used to turn a fan that forces the air to be heated through the system and to the output air duct.

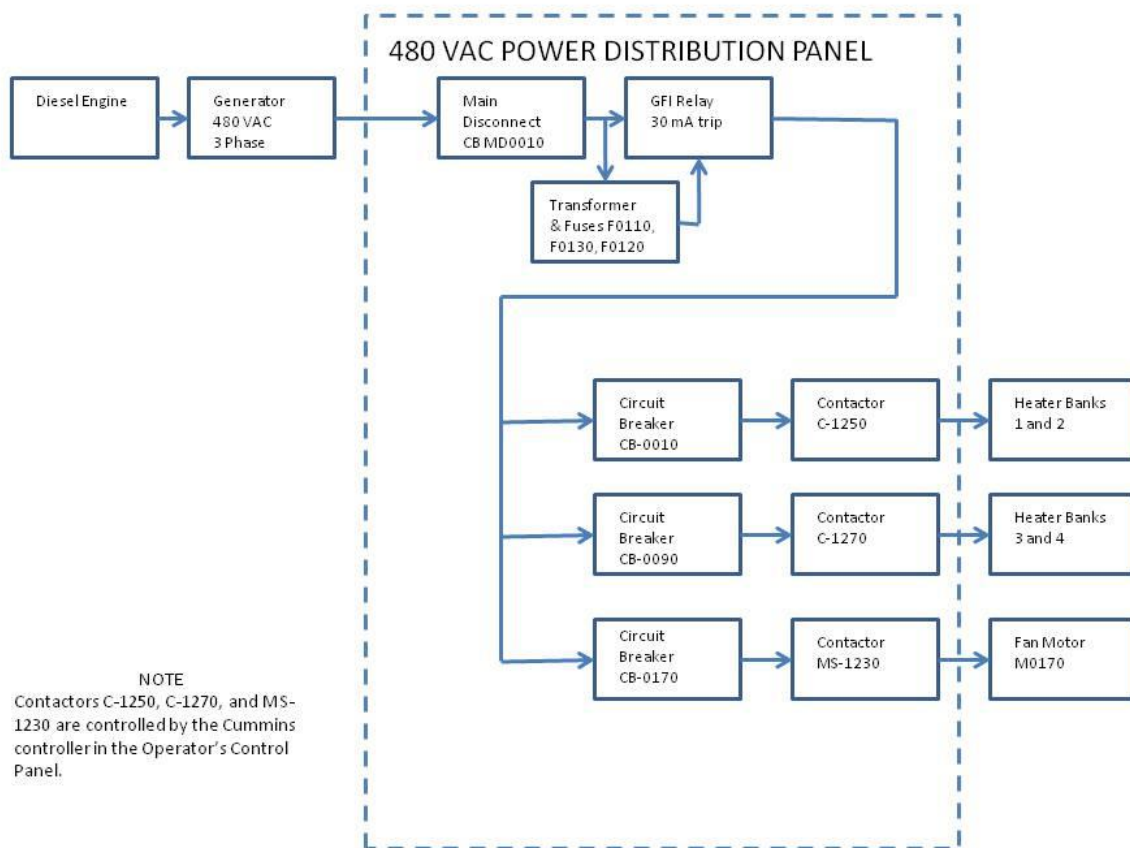


Figure: 480 VAC Power Distribution Panel Theory

THEORY OF OPERATION OF THE OPERATOR'S CONTROL PANEL

The Cummins PCC 1302 Baseboard Controller is the center item of the operator's control panel and shown in the Figure: Operator's Control Panel. It provides monitoring, display and control signals for the Thawzall heater. It communicates with the HMI211 Cummins Display located on the front panel of the operator's control panel and provides status monitoring of the diesel engine and 480 VAC generator. The Cummins PCC 1302 Baseboard Controller also interfaces with the Cummins Aux 101 and 102 System I/O modules to provide interfaces with system sensors and controls.

The 480 VAC generator has sensors on it to monitor the voltage and current of each of the three phases and provides this information to the operator's control panel for display on the HMI211 Cummins Display.

There is a diesel coolant level switch located on the engine that sends a control signal to the operator's control panel to shut down the diesel engine if coolant levels get too low.

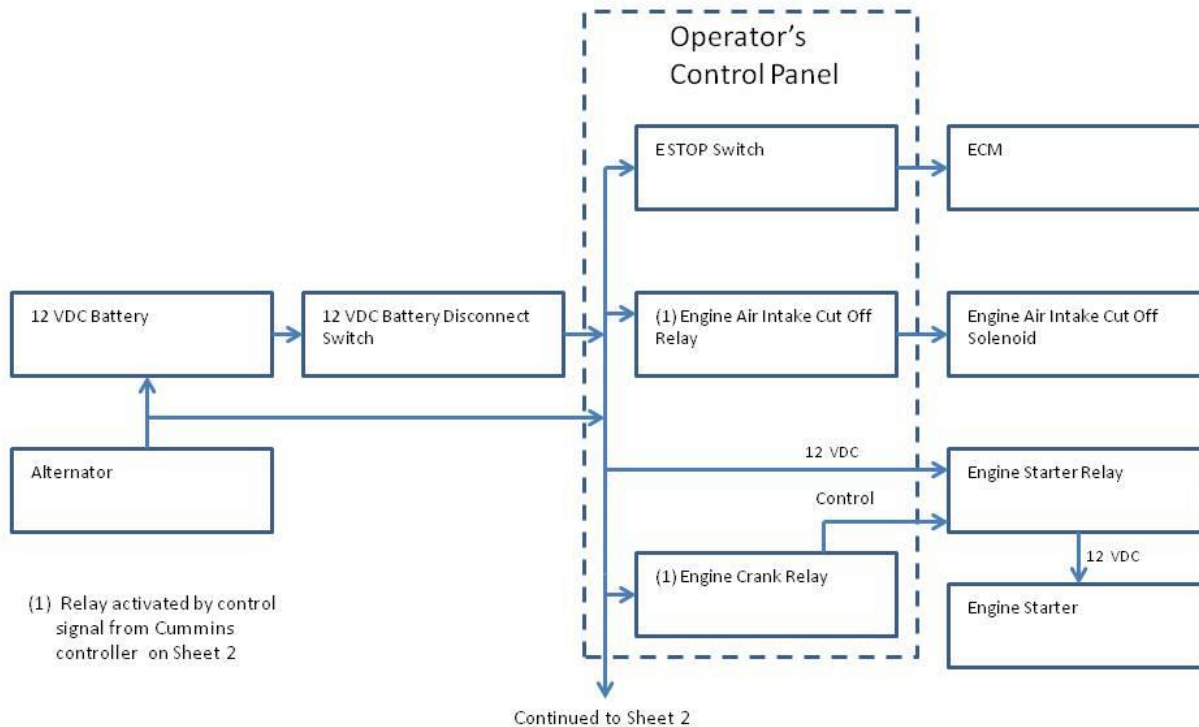


Figure: Operator's Control Panel Theory (sheet 1 of 2)

Control signals from the Cummins PCC 1302 Baseboard Controller are individually sent from the operator's control panel to the 480 VAC power distribution panel to control the activation of the three contactors C1250, C1270, and MS1250 which turn on and off heater banks 1, 2, 3 and 4 along with fan motor M0170.

There is a heater duct over temperature switch in the air duct near the heating banks that senses the air temperature and trips the relay for heater over temperature fan to the "on" state if the air temperature is too high. The "not" function of this relay is in series with the control logic for contactor MS-1230. This function causes this contactor to turn off and that in turn turns off fan motor 0170. When the heater duct over temperature switch is tripped it also prevents the diesel engine from running.

The relay for engine air intake cut off is activated when it verifies that the solenoid for the air intake is engaged. The status of the relay for engine air intake cut off is sensed by the engine air intake limit switch and feed to the operator's control panel for processing. This operation influences the heating operations.

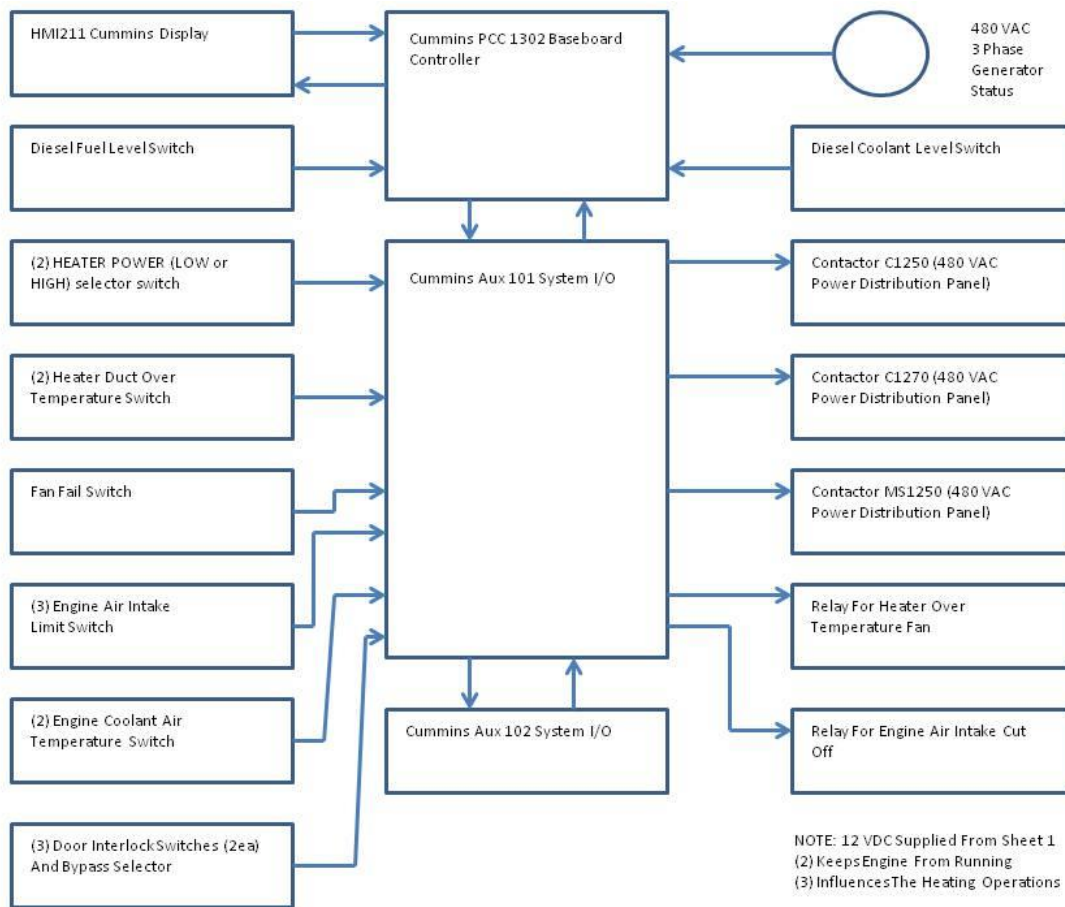


Figure: Operator's Control Panel Theory (sheet 2 of 2)

The fan fail switch provides a signal for the Cummins PCC 1302 Baseboard Controller to know when fan motor M0170 has failed to operate.

The engine coolant air temperature switch located on the diesel engine senses when the coolant temperature is too high and prevents the diesel engine from running when this condition exists.

The diesel fuel level switch senses the fuel level in the diesel fuel tank and feeds this information to the operator's control panel for display.

The HEATER POWER selector switch has two positions. In the LOW position only heater banks 1 and 2 are on. In the HIGH position heater banks 1, 2, 3 and 4 are on.

There are two door interlock switches for access doors on the unit. When one or both doors are open the heaters are prevented from operating. This function can be bypassed by the DOOR INTERLOCK switch.

PRODUCT SAFETY

Keep emergency phone numbers for doctors, ambulance service, hospital and the fire department near your phone.

SAFETY INCLUDES PREVENTITIVE MAINTENANCE

Remove paint and adhesives before welding or heating.

Avoid heating near pressurized pipes and hoses, and near a fuel tank.

Dispose of waste properly. Improperly disposing of waste including grease, and plastics, filters and batteries, threaten the ecology of the environment.

HANDLING BATTERIES SAFELY

Battery gas can explode. Keep sparks and flames away from the battery on the generator on your Thawzall

Never place a metal object across the battery posts

Always disconnect the grounded (-) battery clamp first and re-connect it last

Battery acid is poisonous and can burn skin and eat holes in clothing and cause blindness if splashed into the eyes

Flush contaminated skin with water and baking soda. If acid is swallowed, drink water or milk and get medical attention immediately OR **CONTACT YOUR LOCAL POISON CONTROL CENTER IMMEDIATELY!**

STORING YOUR THAWZALL

To store your Thawzall for the summer months:

Park on level ground and block wheels to prevent accidental movement. Close the fuel tank valve.

SAFETY FEATURES OF THE THAWZALL HEAT MACHINE

The following safety features have been built into your Thawzall to ensure the safest possible use of the machine and convenience that enhances your productivity.

Gages at all vital points

Help to maintain proper operation

Trailer axles with electric brakes and breakaway stop system meet safety codes to include NHTSA, CSA, Transport Canada, UL and IBR

MAINTENANCE

Top level maintenance is provided in this section. For detailed maintenance procedures for your Thawzall H750, please refer to the Cummins ***Operation and Maintenance Manual - QSB4.5 Engine*** supplied with your heater. There you will find a full range of maintenance procedures and maintenance scheduling , including:

- Maintenance Guidelines
- Maintenance Procedures at Daily Interval
- Maintenance Procedures at 250 Hours or 3 Months
- Maintenance Procedures at 500 Hours or 6 Months
- Maintenance Procedures at 1000 Hours or 1 Year
- Maintenance Procedures at 2000 Hours or 2 Years
- Maintenance Procedures at 5000 Hours or 4 Years

AIR CLEANER RESTRICTION MAINTENANCE CHECK

Mechanical Indicator

CAUTION

Never operate the engine without an air filter. Intake air must be filtered to prevent dirt and debris from entering the engine and causing premature wear.

NOTE

Do not remove the felt washer from the indicator. The felt washer absorbs moisture.

A mechanical restriction indicator is available to indicate excessive air restriction through a dry-type air cleaner. This instrument can be mounted in the air cleaner outlet or on the instrument panel. The red flag (1) in the window gradually rises as the cartage loads with dirt. After changing or replacing the cartage, reset the indicator by pushing the reset button (2).

Restriction or vacuum indicators need to be installed as close as possible to the turbocharger air inlet in order to obtain a true indication of restrictions.

Vacuum Indicator

Vacuum switches actuate a warning light on the instrument panel when the air restriction becomes excessive.



Figure: Air Cleaner Restrictor

CHANGING FUEL FILTER (Spin-On Type) MAINTENANCE

Remove

Remove the spin-on fuel filter with fuel wrench, Part Number 3398231.

Install

NOTE: Do not pre-fill an on engine fuel filter

CAUTION

Do not pre-fill an on-engine fuel filter with fuel. This system must be primed after the fuel filter is installed. Pre-filling the fuel filter can result in debris entering the fuel system and damaging fuel system components

Lubricate the O-ring seal with clean lubrication oil.

CAUTION

Use the correct fuel filter. See Cummins/Fleetguard/Nelson filter specifications for the correct part number.

Install the filter on the filter head. Tighten the filter until the gasket contacts the filter head surface.

Tighten the fuel filter an additional $\frac{3}{4}$ turn after contact, or consult the filter manufacturer instructions.

Prime fuel system after filter installation.

Prime

WARNING

The fuel pump high-pressure fuel lines and fuel rail contain very high-pressure fuel. Never loosen any fittings while the engine is running. Personal injury and property damage can result.

To prime the engine use the OEM installed priming device. Typically, a priming pump is installed at or near the pre-filter. See the OEM's instructions for the number of strokes (hand primer) or cycles time (electric priming pump) needed to prime the low pressure system.

NOTE: It is not necessary to vent air from the high pressure system before starting the engine.

NOTE: To prevent damage to the hand pump priming seals, clean the fuel pump head and priming pump with Quick Dry Spray Cleaner, Part Number 3824510, or equivalent, and compressed air prior to priming the fuel system.

Operate the engine and check for leaks.



Fuel Filter

CHANGING LUBRICATING OIL AND FILTERS MAINTENANCE

Drain

WARNING

Some state and federal agencies have determined that used oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

WARNING

To reduce the possibility of personnel injury, avoid direct contact of hot oil with your skin.

Change the lubricating oil and filter at the specified oil change interval. See the maintenance schedule to find the correct change interval for your application.

NOTE: For most engines, use a container that can hold at least 20 liters (21qt) of lubricating oil. Some engines can be equipped with an increased capacity oil pan requiring a container that will hold 28 liters (30 Qt.) of lubricating oil.

Operate the engine until the water temperature reaches 60 ° C (140° F).

Shut off the engine

Remove the oil drain plug. Drain the oil immediately to be sure all the oil and suspended contaminants are removed from the engine.

Remove

Clean the area around the lubricating oil filter head.

Use the oil filter wrench, Par Number 3400158, to remove the filter.

Clean the gasket surface of the filter head.

NOTE: The O-ring can stick on the filter head. Be sure it is removed before installing the new filter.

Install

Use the correct oil filter. See the Cummins/Fleetguard/Nelson specifications for the correct oil filter part number.

CAUTION

The lack of lubrication during the delay until the filter is pumped full of oil at start-up can damage the engine.

Use clean 15W-40 oil to coat the gasket surface of the filter.

Fill the filter with clean 15W-40 oil.

Apply a light film of lubricating oil to the gasket sealing surface before installing the filter.

NOTE: Be careful the no debris is poured into the filter. If using an oil supply with a metallic or plastic seal under the cap, be careful to peel the seal back. Punching the seal with a knife or sharp object can create debris in the oil container.

CAUTION

Mechanical overtightening of the filter can distort the threads or damage the filter sealing element seal.

Install the filter on the oil filter head. Tighten the filter until the gasket contacts the filter head surface.

Tighten $\frac{3}{4}$ turn after the gasket makes contact with the filter head.

Fill

Clean and check the lubricating oil drain plug threats and sealing surface. Use a new sealing washer, if damaged.

Install the lubricating oil pan plug.

Steel Oil Pan Drain Plug Torque			
	Nm		ft-lb
M18	60	Min	44
M22	80	Min	59
Cast Aluminum Oil Pan Drain Plug Torque			
M22	60	Min	44

Fill the engine with clean lubricating oil to the proper level.

Idle the engine to inspect for leaks at the drain plug and , if replaced, the oil filter seal.

NOTE: Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut off the engine immediately to avoid engine damage. Confirm that the correct oil level is in the oil pan.

Shut off the engine. Wait approximately 5 minutes to let the oil drain from the upper parts of the engine. Check the level again. Add oil as necessary to bring the oil level to the H (high) mark on the dipstick.



Figure: Drain Plug Location

Figure: Oil Fill Location



OIL LEVEL MAINTENANCE CHECK

Maintenance Check

CAUTION

Never operate the engine with oil level below the L level or above the H level. Poor engine performance or engine damage can occur.

The engine must be level when checking the oil level to make sure the measurement is correct.

Shut off the engine for an accurate reading.

Wait at least 15 minutes after shutting off the engine to check the oil level. This allows time for the oil to drain into the oil pan.



Figure: Dipstick Location

AIR FILTER FOR HEATER AIR DUCT MAINTENANCE CHANGE

Remove filter from filter enclosure

Inspect filter for dirt and if dirty replace.



Figure: Heating Duct Air Filter Location

BATTERY MAINTENANCE CHECK

Make sure battery terminals are clean and a secure connection.

Check Battery with a battery tester.



Figure: Top View of Battery

TROUBLESHOOTING

Use the chart shown below when troubleshooting codes on the HMI display on the Operator's Control Panel.

FAULT AND STATUS CODES FOR CUMMINS DISPLAY				
Code	Lamp	Active State	Input	Output
234	Shutdown	See Cummins Manual	Internally Generated	K6 (Output)
1468	None	See Cummins Manual	Internally Generated	K1 and K9 (Outputs 1 & 9)
1483	None	See Cummins Manual	Internally Generated	K10 (Output 10)
1541	None	See Cummins Manual	Internally Generated	K11 (Output 11)
2619	Diagnostic	Low	Input 1	K3 (Output 3)
2621	Diagnostic	Low	Input 2	K5 (Output 5)
2622	Diagnostic	Low	Input 3	Internally, Kills Engine
2623	Diagnostic	High	Input 4	K2 (Output 2)
2624	Diagnostic	High	Input 5	Internally, Kills Engine
2625	Diagnostic	Low	Input 6	K4 (Output 4), currently unused

The following table has been developed to address fault codes presented by the HMI. It displays the settings in the HMI to allow understanding of the actual function of the inputs and outputs as an aid in determining expected conditions on the H750 and their affect on fault conditions..

HMI FAULT CODES AND SETTINGS					Active Dates:	11/21/2012 - present	
Genset Service Menu							
Customer I/O Settings 1302 Inputs							
Input 1	<u>Not In Use!</u>			Disabled	Input Points:	TB1-12 to TB1-13	This input is not presently in use.
	Type:	Warning	Input Normal State:		N/A		
	Text:	Fault Input #1	Input Normal Voltage:		5 VDC		
	Active State Select:	Active Close	Input Active Voltage:		0 VDC		
	Fault Code:						
Input 2	<u>Not In Use!</u>			Disabled	Input Points:	TB1-14 to TB1-13	This input is not presently in use.
	Type:	Warning	Input Normal State:		N/A		
	Text:	Fault Input #2	Input Normal Voltage:		5 VDC		
	Active State Select:	Active Close	Input Active Voltage:		0 VDC		
	Fault Code:	1541					
Input 3				Disabled	Input Points:	J20-17 to J20-5	coolant level switch.
	Type:	Warning	Input Normal State:		open {cloosed when empty}		

	Text:	LOW COOLANT LEVEL		Input Normal Voltage:	5 VDC	
	Active State Select:	Active Close		Input Active Voltage:	0 VDC	
	Fault Code:					
Input 4				Input Points:	J20-18, J20-21, J20-6	Fuel Level Sensor.
Type:		Warning	<i>Disabled</i>			
Text:		LOW FUEL LEVEL				
Active State Select:		Active Close				
Fault Code:						

Customer I/O Settings					
1302 Outputs					
Output 1 Map			Output Points:	TB1-6 to TB7	This Output is not presently in use.
			Output Normal State:	N.O.	
	Fault Code:	Fault Output #1			
Output 2 Map			Output Points:	TB1-8 to TB9	This Output is not presently in use.
			Output Normal State:	N.O.	
	Fault Code:	Fault Output #2			

I/O Moodule Settings						
Aux 101 Inputs						

Input 1			Input Points:	J11-2 to J11-1 (ref)	HEATER POWER, LOW/HIGH selector switch . Low = 1 , High = 0 Referenced as 2619 on drawings.
	Text:	HIGH HEAT	Input Normal State:	Open	
			Input Normal Voltage:	5 VDC	
			Input Active Voltage:	0 VDC	
Input 2			Input Points:	J11-4 to J11-3 (ref)	FAN FAIL O/L contact OR HEAT DUCT HIGH TEMP switch. Both OK=1, either fail=0. Referenced as 2621 on drawings.
	Text:	DUCT OT or FAN O/L	Input Normal State:	Open	
		Inactive	Input Normal Voltage:	5 VDC	
			Input Active Voltage:	0 VDC	
Input 3			Input Points:	J11-6 to J11-5 (ref)	AIR INTAKE CUTOFF VALVE ACTUATED. N.O. held closed. Referenced as 2622 on drawings.
	Text:	AIR VALVE TRIP	Input Normal State:	Closed	
		Inactive	Input Normal Voltage:	0 VDC	
			Input Active Voltage:	5 VDC	
Input 4			Input Points:	J11-8 to J11-7 (ref)	COOLENT TEMP. N.O., active above 90° F. Referenced as 2623 on drawings.
	Text:	READY FOR LOAD (at Temp)	Input Normal State:	Open	
		Inactive	Input Normal Voltage:	5 VDC	

				Input Active Voltage:	0 VDC	
Input 5	Text:	I/O Input 5		Input Points:	J11-10 to J11-9 (ref)	Open Input. Referenced as 2625 on drawings.
				Input Normal State:	Closed Guard Doors and open Interlock	
		Inactive		Input Normal Voltage:	0 VDC	
				Input Active Voltage:	5 VDC	
Input 6	Text:	I/O Input 6		Input Points:	J11-12 to J11-11 (ref)	Open Input. Referenced as 2625 on drawings.
				Input Normal State:		
		Inactive		Input Normal Voltage:		
				Input Active Voltage:		
Input 7	Text:	I/O Input 7		Input Points:	J11-14 to J11-13 (ref)	Open Input. Referenced as 2626 on drawings.
				Input Normal State:		
		Inactive		Input Normal Voltage:		
				Input Active Voltage:		
Input 8						

	Text:	I/O Input 8	Input Points: Input Normal State: Input Normal Voltage: Input Active Voltage:	J11-16 to J11-15 (ref)	Open Input. Referenced as 2627 on drawings.

Input 9	N/A				Currently not in use
	Text:	I/O Input 9			
		Inactive			

Input 10	N/A				Currently not in use
	Text:	I/O Input 10			
		Inactive			

Input 11	N/A				Currently not in use
	Text:	I/O Input 11			
		Inactive			

Input 12	N/A				Currently not in use
	Text:	I/O Input 12			
		Inactive			

I/O Moodule Settings					
Aux 101 Outputs					
Output 1			Output Point:	J2-1 N.O. Contact	FAN ON: K1 Relay. If K1 is active (Fan On, MS1230) J2-1 and J2-2 are in contact (12 VDC between J2-
			Output Point:	J2-2 Common Contact	

	FAULT NO:	1468		Output Point:	J2-3 N.C. Contact	1 and J2-3 and 0VDC reference).
Output 2			{Blinking}	Output Point:	J2-4 N.O. Contact	HEATING STEP 1: K2 Relay. If K2 is active (Heating Step 1, C1250) J2-4 and J2-5 are in contact (12 VDC between J2- 4 and J2-5 and 0VDC reference). Also Starts Delay timer for High heat.
	FAULT NO:	Ready for Load 2623		Output Point:	J2-5 Common Contact	
				Output Point:	J2-6 N.C. Contact	
Output 3			{Blinking}	Output Point:	J2-7 N.O. Contact	HEATING STEP 2: K3 Relay. If K3 is active (Heating Step 2, C1270) J2-7 and J2-8 are in contact (12 VDC between J2- 7 and J2-8 and 0VDC reference).
	FAULT NO:	High Heat 2619		Output Point:	J2-8 Common Contact	
				Output Point:	J2-9 N.C. Contact	
Output 4				Output Point:	J2-10 N.O. Contact	Currently not in use States fault 2625 on

		FAULT NO:		Output Point:	J2-11 Common Contact	drawings. Output 4 is not presently used.
				Output Point:	J2-12 N.C. Contact	
Output 5				Output Point:	J3-1 N.O. Contact	HEAT OVER_TEMP FAN FAILURE: K5 Relay. If K5 is active (Heat over temp fan failure, C1310) J3-1 and J3-2 are in contact (12 VDC between J3- 1 and J3-2 and 0VDC reference).
				Output Point:	J3-2 Common Contact	
				Output Point:	J3-3 N.C. Contact	
		FAULT NO 1:	2621			
		FAULT NO 2:	0			
		FAULT NO 3:	0			
Output 6				Output Point:	J3-4 N.O. Contact	AIR INTAKE CUT_OFF (KILL ENGINE): K6 Relay. If K6 is active (AIR INTAKE CUT_OFF C1330) J3-4 and J3-5 are in contact (12 VDC between J3- 4 and J3-5 and 0VDC reference).
				Output Point:	J3-5 Common Contact	
				Output Point:	J3-6 N.C. Contact	
		FAULT NO:	234			

Output 7			Output Point:	J4-1 N.O. Contact	Currently not in use
			Output Point:	J4-2 Common Contact	
			Output Point:	J4-3 N.C. Contact	
	FAULT NO:	0			

Output 8			Output Point:	J4-4 N.O. Contact	Currently not in use
			Output Point:	J4-5 Common Contact	
			Output Point:	J4-6 N.C. Contact	
	FAULT NO:	0			

Output 9			Output Point:		Currently not in use
			Output Point:		
			Output Point:		
	FAULT NO:	0			

Output 10			Output Point:		Currently not in use
			Output Point:		
			Output Point:		
	FAULT NO:	0			

Output 11			Output Point:		Currently not in use
			Output Point:		
			Output Point:		
	FAULT NO:	1541			

Output 12			Output Point:		Currently not in use
			Output Point:		
			Output Point:		
	FAULT NO:	0			

Output 13			Output Point:		Currently not in use
			Output Point:		
			Output Point:		
	FAULT NO:	0			
Output 14			Output Point:		Currently not in use
			Output Point:		
			Output Point:		
	FAULT NO:	0			
Output 15			Output Point:		Currently not in use
			Output Point:		
			Output Point:		
	FAULT NO:	0			
Output 16			Output Point:		Currently not in use
			Output Point:		
			Output Point:		
	FAULT NO:	0			

WIRING SCHEMATICS

82043 480 VAC Power Distribution Panel

82046 Control Interconnections

82313 Operator's Control Panel 1 of 2 and 2 of 2

82391 Power Interconnections



LIMITED WARRANTY

GENERAL:

THAWZALL, LLC hereby extends to the original purchaser of its THAWZALL ("Ground Defrosting, Thawing, Temporary Heat or Concrete Curing Products") a warranty against defects in materials and workmanship for the two year time period indicated below.

The warranty is only valid on "Ground Defrosting, Thawing, Temporary Heat or Concrete Curing Products" purchased and used in accordance with placards and instructions (e.g. Operators Manuals) provided by Thawzall, LLC. This warranty applies only to the original purchaser and is subject to the terms and conditions set forth below.

THAWZALL, LLC will repair or replace (at its sole option) a ground defrosting, thawing, temporary heat or concrete curing product (or component thereof) if it fails to conform to this warranty. In the event a ground defrosting, thawing, temporary heat or concrete curing product is to be repaired pursuant to this warranty, such repair work will be performed by THAWZALL, LLC or at its direction.

WARRANTY PERIOD:

The warranty relating to workmanship, materials and labor on THAWZALL ground defrosting, temporary heat or concrete curing products extends for two (2) years from the date of original invoice.

WARRANTY POLICY:

Within 5 working days after receiving a properly completed warranty claim Thawzall, LLC will make a determination as to the validity of the claim and, if the claim is deemed to have merit, will:

- Issue credit for 50% of the amount entitled based upon current pricing and warranty labor reimbursement rates, and

- Issue an RMA for the parts in question, If the RA parts are received with 30 days of the RMA date, Thawzall, LLC will issue credit for the balance of the amount entitled.

- RMA Parts not returned within 30 days will no longer be eligible for any credit.

WARRANTY PROCEDURE:

RMA – (Return Material Authorization): To ensure processing of warranty claim, a Return Merchandise Authorization ("RMA") must be obtained and prominently shown on correspondence and packages. To obtain an RA, call 888.757.3545 (U.S. Central Time) or [E-mail warranty@thawzall.com](mailto:warranty@thawzall.com). Parts must be returned within 30 days of an RA being issued.

Freight Charges and Handling Fees: The purchaser is responsible for shipping charges on any items returned with a valid RA number.

Proof of Purchase: Proof of purchase (invoice number and date of invoice) identifying the model number and serial number of ground defrosting, thawing, temporary heat or concrete curing products must accompany warranty claim.

WARRANTY LIMITATIONS:

Thawzall Ground Defrosting, Thawing, Temporary Heat or Concrete Curing products must be installed (where applicable), operated and maintained in accordance with all instructions provided by Thawzall, LLC. Failure to follow our installation (where applicable), operating or maintenance procedures and/or use of unauthorized parts may void this

warranty.

Purchasers and Users are responsible for the suitability of the products for their application.

This warranty does not apply to:

Repairs or replacements necessitated by any cause beyond the control of THAWZALL, LLC including, but not limited to, any malfunction, defect or failure caused by or resulting from unauthorized service or parts; installation (where applicable), operating or maintenance contrary to furnished instructions; local water conditions, handling, shipping or transit accidents; modifications or repair by the user; abuse; misuse; neglect; accident; incorrect power line voltage; power line surge; lightning damage; or fire, flood, or other Acts of God.

Repair or replacement in the ordinary course of expendable ground defrosting, thawing, temporary heat or concrete curing product part

Elements and controls whose damage or failure is attributable to corrosion, scale, or dirt accumulations or to low water conditions

Thawzall, LLC is not liable for labor and other costs incurred in removal, reinstallation, or unauthorized repair of the Ground Defrosting, Thawing, Temporary Heat or Concrete Curing product or for damages of any type whatsoever including incidental or consequential damages.

There are no warranties which extend beyond the description contained herein and specifically liability for any breach of any implied warranty of merchantability or fitness for a purpose is excluded. The duration of any warranties which may be implied by law notwithstanding the previous sentence (including the warranties of merchantability and fitness) is limited to the term of this warranty. In no event shall Thawzall, LLC be liable for special, incidental or consequential damages arising from ownership or use of any Ground Defrosting, Thawing, Temporary Heat, or Concrete Curing product, or for any delay in the performance of its obligations under this warranty due to causes beyond its control. Some states do not allow limitations on how long an implied warranty lasts and/or do not allow the exclusion or limitation of consequential damages, so the above limitations and exclusion may not apply to you. This warranty gives you specific legal rights. You may have other rights, which vary from state to state.

This warranty set forth herein is in lieu of all other expressed or implied warranties. THAWZALL, LLC does not assume or authorize any party to assume for it any other obligation or liability.

THAWZALL, LLC

104 Donovan Drive

PO Box 158

Alexandria, MN 56308 USA

TF 888.757.3545

T 320.759.1588

F 320.634.4563

Email: warranty@thawzall.com

Website: www.thawzall.com

How to File a Warranty Claim

The warranty on your Thawzall covers parts and labor for two years from the date of purchase.

In the event of a failure which is covered in the Warranty Statement on the preceding pages:

- Access the warranty claim form online at: <http://www.thawzall.com/warranty.asp?p=1.1.9>
- Or, fill out claim form on the next page; fill it out completely and send it to Thawzall via USPS Mail, Fax or E-mail
- Thawzall will determine the validity of the claim and issue a 50% credit to you for all valid claims within 5 days
- Please call 888.757.3545 or e-mail us at warranty@Thawzall.com to request a Return Material Authorization (RMA) number
- Thawzall will issue you a RMA number for the failed parts.
- To collect the balance of the credit owed, you MUST return the failed parts with the RMA form to Thawzall within 30 days
- When the failed parts are returned to Thawzall, you will receive the remaining credit.

WARRANTY CLAIM FORM

Thawzall warranties hydronic equipment for two years and flameless heaters for one year from the original invoice date.

RMA (Return Material Authorization) number is required for parts re- turn. RMA must be clearly marked on the outside of the returned pack- age. Parts must be returned within 30 days after RMA is issued



THAWZALL, LLC

104 Donovan Drive

Alexandria, MN 56308

USA O 320.759.1588

F 320.759.1583

T 888.757.3545

www.thawzall.com

TODAY'S DATE	
YOUR NAME	
YOUR TITLE	
COMPANY	
E-MAIL	
PHONE	
FAX	
CELL	

MODEL NUMBER	
VIN NUMBER	
DATE OF ORIGINAL	

Describe problem in detail:

How was the problem resolved?

LABOR HOURS

HOURLY LABOR RATE

IF PARTS WERE ORDERED FROM THAWZALL RECORD SALES